

Core Pure Mathematics 2

Notes · Edexcel A Level Further Mathematics — Core Pure 2 · Chapter 8: Modelling with Differential Equations

Chapter 8: Modelling with Differential Equations

Why this matters

Chapter 7 gave you the machinery to *solve* differential equations. This final chapter is about the equally important skill of *setting them up* from a real-world description — translating words like "the rate of change of temperature is proportional to the difference between the object's temperature and the surrounding temperature" into a precise equation — and then interpreting the solution back in context. This is where further maths differential equations meet real modelling, and it's a favourite source of extended exam questions.

Concept: translating a rate-of-change statement into an equation

The key phrase to watch for is "rate of change of [quantity] is proportional to ...", which always translates to a derivative set equal to a constant times an expression:

"Rate of change of x is proportional to y " $\rightarrow \frac{dx}{dt} = ky$, for some constant k (which may be positive, for growth, or negative, for decay — the sign is often determined by the context, or left as a constant to be found from data).

Always define your variables and state units clearly at the start of a modelling answer — this is often an explicit mark in Edexcel mark schemes.

Worked example 1 — Simple proportional growth (setting up and solving)

WORKED EXAMPLE 1

Question. The number of bacteria, N , in a colony is growing at a rate proportional to the number present at time t hours. Initially there are 200 bacteria, and after 3 hours there are 600. Form and solve a differential equation for N in terms of t , and find the number of bacteria after 6 hours.

Step 1. Translate the statement into a differential equation: **(M1)** $\frac{dN}{dt} = kN$, for some constant $k > 0$

Step 2. Separate variables and integrate: **(M1)** $\int \frac{1}{N} dN = \int k dt \Rightarrow \ln N = kt + c$

Step 3. Solve for N : $N = e^{(kt+c)} = Ae^{(kt)}$, where $A = e^c$. **(M1)**

Step 4. Use $N = 200$ at $t = 0$ to find A : $200 = Ae^0 = A$, so $A = 200$. **(A1)**

Step 5. Use $N = 600$ at $t = 3$ to find k : **(M1)** $600 = 200e^{(3k)} \Rightarrow e^{(3k)} = 3 \Rightarrow 3k = \ln 3 \Rightarrow k = (\ln 3)/3$

Step 6. Substitute $t = 6$ into $N = 200e^{(kt)}$: **(M1)** $N = 200e^{(6 \cdot (\ln 3)/3)} = 200e^{(2 \ln 3)} = 200 \cdot 3^2 = 200 \times 9 = 1800$

Answer: $N = 200e^{((\ln 3)t/3)}$; after 6 hours there are 1800 bacteria. **(A1)**

EXAM TIP

Notice the shortcut in Step 6: since 6 hours is exactly double the 3-hour period in which the population tripled, the population must triple again — giving $200 \times 3 \times 3 = 1800$ without needing k explicitly. This kind of "doubling/tripling time" reasoning is a powerful modelling instinct worth building early, and it also gives you an instant way to check your k -value is right.

Concept: Newton's law of cooling

A classic modelling context: the rate of change of temperature of an object is proportional to the difference between its temperature and the temperature of its surroundings. If θ is the object's temperature at time t and θ_s is the (constant) surrounding temperature:

$$d\theta/dt = -k(\theta - \theta_s), \quad k > 0$$

The minus sign is essential: it ensures the object cools when $\theta > \theta_s$ ($d\theta/dt$ negative) and warms when $\theta < \theta_s$ ($d\theta/dt$ positive) — either way, θ moves *towards* θ_s over time.

Worked example 2 — Newton's law of cooling
WORKED EXAMPLE 2

Question. A cup of tea is made with boiling water at 100°C in a room at a constant 20°C . Its temperature θ ($^\circ\text{C}$) at time t minutes satisfies $d\theta/dt = -k(\theta - 20)$. After 5 minutes the tea has cooled to 70°C . Find θ in terms of t , and find how long it takes to cool to 40°C .

Step 1. Separate variables: **(M1)** $\int 1/(\theta - 20) d\theta = \int -k dt \Rightarrow \ln|\theta - 20| = -kt + c$

Step 2. Solve for θ : $\theta - 20 = Ae^{(-kt)}$, where $A = e^c$, so $\theta = 20 + Ae^{(-kt)}$. **(M1)**

Step 3. Use $\theta = 100$ at $t = 0$: $100 = 20 + A \Rightarrow A = 80$. **(A1)**

Step 4. Use $\theta = 70$ at $t = 5$ to find k : **(M1)** $70 = 20 + 80e^{(-5k)} \Rightarrow 50 = 80e^{(-5k)} \Rightarrow e^{(-5k)} = 5/8 \Rightarrow -5k = \ln(5/8) \Rightarrow k = -(1/5)\ln(5/8) = (1/5)\ln(8/5)$

Step 5. Substitute $\theta = 40$ and the value of k , and solve for t : **(M1)(M1)** $40 = 20 + 80e^{(-kt)} \Rightarrow 20 = 80e^{(-kt)} \Rightarrow e^{(-kt)} = 1/4 \Rightarrow -kt = \ln(1/4) \Rightarrow t = \ln 4 / k$

Step 6. Substitute the value of k found in Step 4: **(A1)** $t = \ln 4 / [(1/5)\ln(8/5)] = 5 \ln 4 / \ln(8/5) \approx 5(1.386)/0.470 \approx 14.7$ minutes

Answer: $\theta = 20 + 80e^{(-kt)}$ with $k = (1/5)\ln(8/5)$; the tea reaches 40°C after approximately 14.7 minutes.

EXAM TIP

Check the sign of k makes physical sense at the point you find it: k should come out positive here (since θ is decreasing towards 20), so if your algebra gives a negative k , that's an immediate signal to check the $-k(\theta - \theta_s)$ setup rather than trusting the arithmetic. Building this "does the sign make physical sense" habit catches a large fraction of modelling errors before they propagate.

Concept: forming and solving a second-order model

Some real contexts (springs, oscillations, certain population models with resistance) lead naturally to second-order differential equations, solved exactly as in Chapter 7 but with the constants and variables given physical meaning. Often the question gives you the physical relationship (e.g. "the restoring force is proportional to displacement") and expects you to write down and solve the resulting equation, then interpret A, B, or the type of solution (oscillatory, exponential decay, etc.) in context.

If the context includes an external driving force as well as a restoring force (e.g. "a periodic force $F \cos(\omega t)$ is also applied"), the right-hand side of the equation is no longer zero, and you're solving a **non-homogeneous** equation exactly as in the last section of Chapter 7: find the CF (the natural oscillation) and a PI (the response driven by the forcing term), then combine them and apply the initial conditions to the sum.

Worked example 3 — Setting up and solving a second-order model

WORKED EXAMPLE 3

Question. A particle P moves on the x-axis such that its acceleration is always directed towards the origin O and has magnitude $9x$, where x is its displacement from O. Initially, P is at $x = 2$ with velocity 0. Form a differential equation for x in terms of t , solve it, and describe the resulting motion.

Step 1. Since the acceleration is directed *towards* O with magnitude $9x$, and acceleration = d^2x/dt^2 , we get (the restoring nature gives a negative sign): **(M1)** — **correct sign for a restoring acceleration** $d^2x/dt^2 = -9x$, i.e. $d^2x/dt^2 + 9x = 0$

Step 2. Auxiliary equation: $m^2 + 9 = 0 \implies m^2 = -9 \implies m = \pm 3i$ (complex roots with $p = 0$, $q = 3$). **(M1)(A1)**

Step 3. Write the general solution: **(A1)** $x = e^{(0 \cdot t)}(A \cos 3t + B \sin 3t) = A \cos 3t + B \sin 3t$

Step 4. Apply $x = 2$ at $t = 0$: $A = 2$. **(M1)**

Step 5. Differentiate: $dx/dt = -3A \sin 3t + 3B \cos 3t$. Apply $dx/dt = 0$ at $t = 0$: $3B = 0 \implies B = 0$. **(M1)**

Answer: $x = 2 \cos 3t$. This describes **simple harmonic motion**: P oscillates between $x = -2$ and $x = 2$ with period $2\pi/3$, always returning to and passing through the origin, since the solution is a pure cosine with no exponential growth or decay term. **(A1)**

EXAM TIP

This is a natural bridge to Further Mechanics's treatment of simple harmonic motion (SHM) — notice the connection: whenever the auxiliary equation gives purely imaginary roots $\pm qi$ ($p = 0$), the motion is undamped oscillation, exactly matching the SHM equation $d^2x/dt^2 = -\omega^2x$ that you may already have met. If $p \neq 0$ as well, the $e^{(px)}$ factor represents damping (decaying oscillation if $p < 0$) — a great preview of damped SHM.

Concept: interpreting the solution in context

A modelling question is rarely finished once you've solved the equation — Edexcel routinely asks you to comment on the model. Common things to check or discuss:

- **Long-term behaviour:** what happens as $t \rightarrow \infty$? (e.g. does a population stabilise, grow without bound, or decay to zero?)
- **Validity of the model:** is exponential growth realistic forever (e.g. for a bacteria population limited by food/space)? Often the answer is "no — the model will break down for large t " and a more realistic model (e.g. one with a limiting/carrying capacity) would be needed.
- **Physical restrictions:** are there sensible restrictions on t or on the variable (e.g. temperature cannot go below room temperature, so $\theta > \theta_s$ always in a cooling model — check the solution respects this).

Worked example 4 — Commenting on the validity of a model

WORKED EXAMPLE 4

Question. A population P of rabbits is modelled by $dP/dt = 0.1P$, with $P = 50$ at $t = 0$ (t in years). Solve the equation, and comment on one limitation of this model for large t .

Step 1. Separate and solve as before: **(M1)(A1)** $P = 50e^{(0.1t)}$

Step 2. Comment on validity: **(B1)**

Answer: $P = 50e^{(0.1t)}$. As $t \rightarrow \infty$, $P \rightarrow \infty$, but in reality a rabbit population cannot grow without bound forever — food, space, and predation would eventually limit growth, so the exponential model is only realistic for relatively small t (early-stage growth), and a model with a limiting capacity (e.g. a logistic-type equation) would be more realistic for large t .

Chapter summary

Idea	Key point
Setting up	"Rate of change of x proportional to ..." $\rightarrow dx/dt = k \times (...)$, define variables and units clearly
Newton's law of cooling	$d\theta/dt = -k(\theta - \theta_s)$; minus sign ensures θ moves towards θ_s
Second-order models	Physical description $\rightarrow$ differential equation $\rightarrow$ auxiliary equation $\rightarrow$ solve as in Chapter 7
Interpreting solutions	Comment on long-term behaviour, realism of the model, and physical restrictions

Common exam mistakes

- Getting the sign wrong when setting up a proportionality (e.g. missing the minus sign in Newton's law of cooling, or in a "restoring towards origin" acceleration).

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- Not defining variables and their units clearly at the start of a modelling answer, losing an easily-available mark.
 - Solving correctly but forgetting to answer the final "interpret" or "comment on the model" part of the question, which frequently carries dedicated marks.
 - Substituting boundary conditions into the wrong form of the general solution (e.g. forgetting to differentiate before applying a velocity/rate condition).
 - Giving purely numerical answers to "describe the motion" questions instead of identifying the type of behaviour (e.g. exponential decay, growth, or oscillation, and stating amplitude/period where relevant).

You should now be able to

- Translate a real-world rate-of-change description into a first-order or second-order differential equation.
- Solve models such as exponential growth/decay and Newton's law of cooling, and use given data to find unknown constants.
- Set up and solve second-order differential equations arising from physical contexts, including recognising simple harmonic motion from complex conjugate roots with zero real part.
- Interpret a solution in context, including commenting on the long-term behaviour and realism of the model.